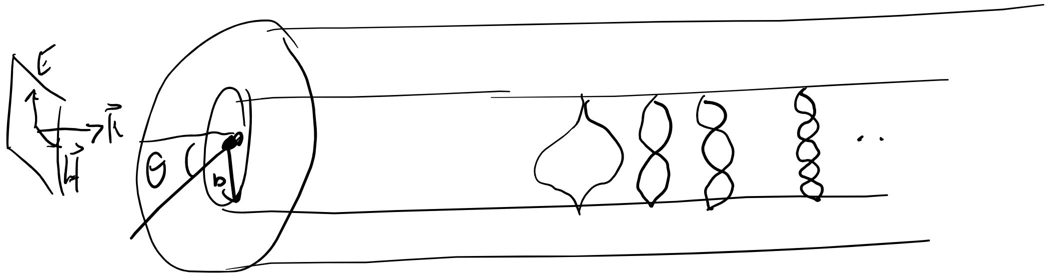
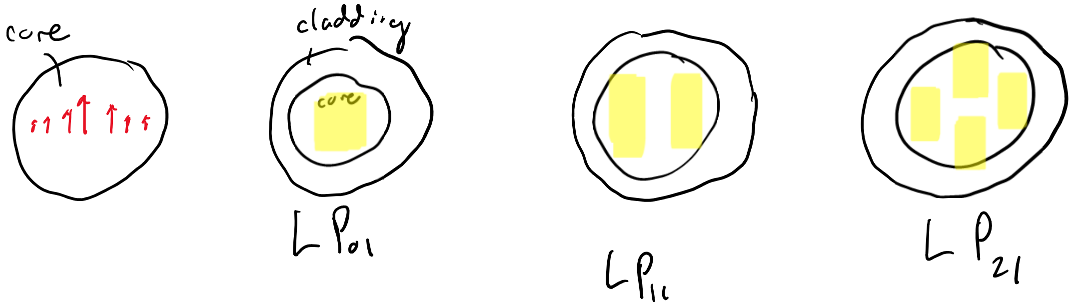
Lecture 31



$\theta > \theta_c$; T.I.R

$$\vec{E}_{linear\text{-}polarized} = E_{lm}(b,\theta)\, e^{j(\omega t - (\beta_{lm})z)}$$

Linear Polarized Modes



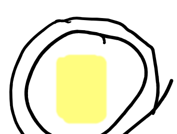
V-parameters; can define the no. of these modes

$$V = \frac{2\pi b}{\lambda_0}\left((n_1)^2-(n_2)^2\right)^{1/2}$$; b is radius of fiber

$$\Delta = \frac{n_1-n_2}{n_1}$$; Δ the normalized index of refraction

$$V = \frac{2\pi b}{\lambda_0}\left(2(n_1)(n)\Delta\right)^{1/2}$$; $n = \frac{n_1+n_2}{2}$ ⇒ average index of refraction

$\boxed{V \le 2.405}$ ⇒ only have <u>one</u> mode supported in the fiber



\# of modes $\boxed{M \approx \frac{V^2}{2}}$ → Round down

$$\lambda_c \text{(cuttoff wavelength)} = \frac{2\pi b}{2.405}\left((n_1)^2 - (n_2)^2\right)^{1/2}$$

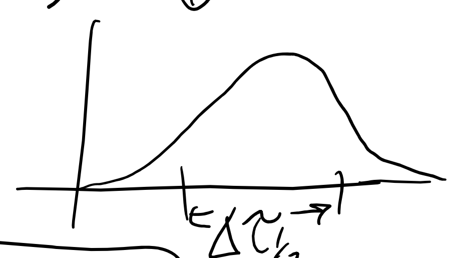
transition b/w a single mode operation and a multimode operation

* sensing applications → single mode

* comms. → multimode.

Optical Communications

- bit Rate capacity (B) $\frac{\text{bits}}{\text{sec}}$
- Disperion speed $(\Delta\tau_{1/2})$
  (bane of all fibers)

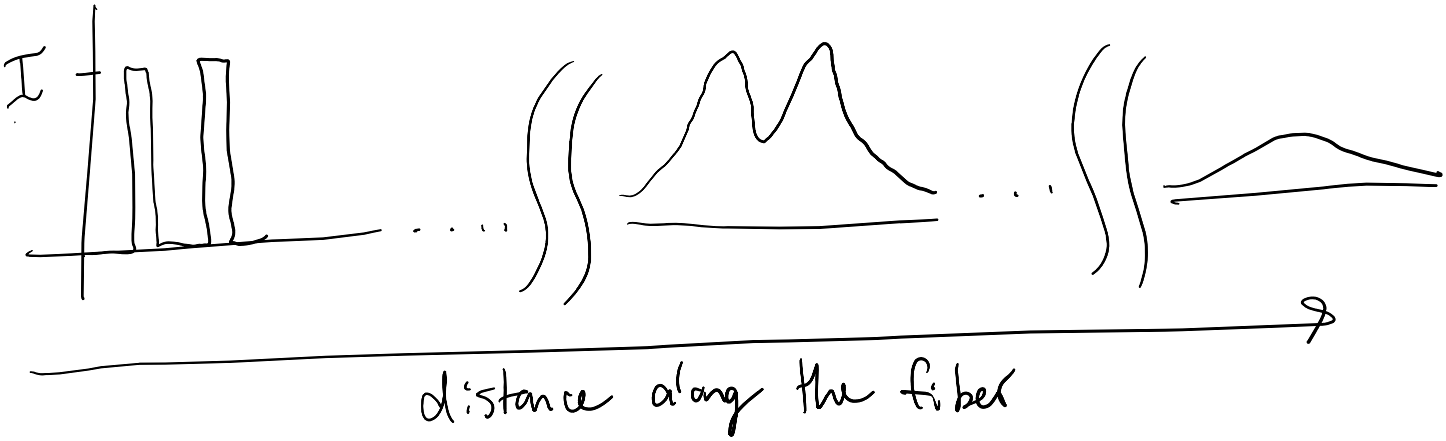


$$\boxed{\Delta\tau_{1/2} = \tau_{RX} - \tau_{TX}}$$

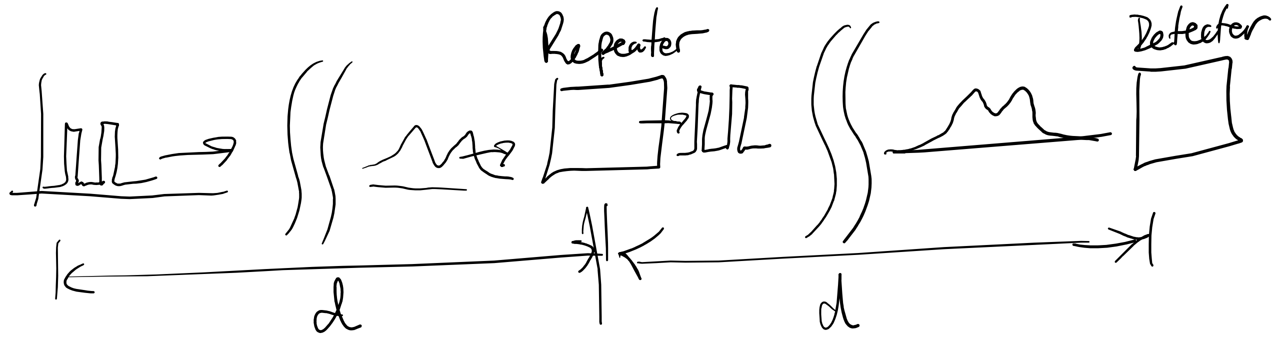
$$\boxed{B \approx \frac{1}{2(\Delta\tau_{1/2})}}$$

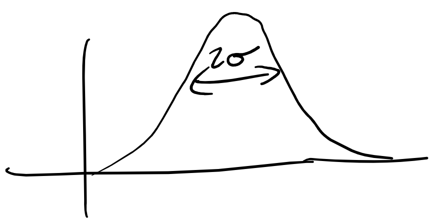
Dispersion in a fiber (comms)

Input to fiber



In a Real system





σ is the standard deviation

$$\sigma \approx 0.425(\Delta\tau_{1/2})$$

$$\boxed{2\sigma = \Delta\tau_{RMS}}$$

length of fiber

$$B \approx \frac{1}{4\sigma} \rightarrow (B)(L) = \frac{L}{4\sigma}$$

$$\boxed{B = \frac{1}{4|D_{CH}|\sigma_\lambda}}$$

(D_{CH} = chromatic dispersion coefficient)

In a fiber there are 2-types of dispersion

- Dispersion in the core (D_{core})
- Dispersion in the core and cladding boundary ($D_{waveguide}$) (due to the evanescent wave in the cladding)

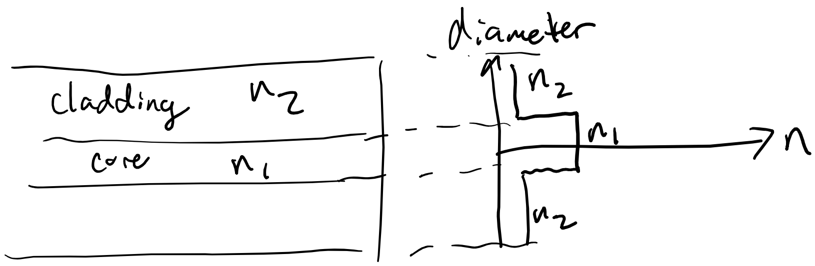
$$\boxed{|D_{CH}| = D_{core} + D_{waveguide}}$$

*

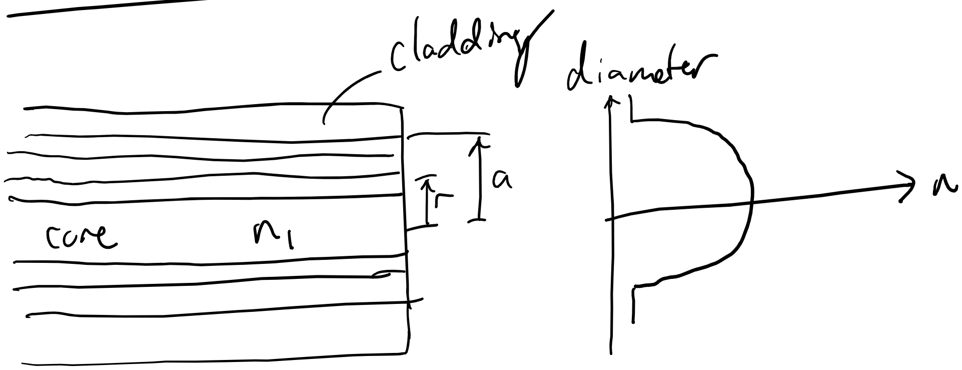
$$\boxed{\frac{\Delta \tau_{RMS}}{L} = |D_{CH}| \Delta \lambda}$$

"Better" ways of designing fibers

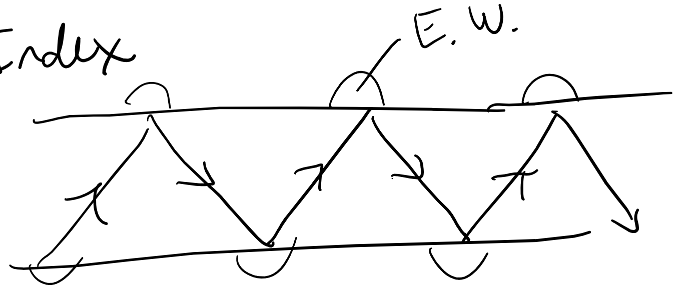
- Step-Index (support single mode)



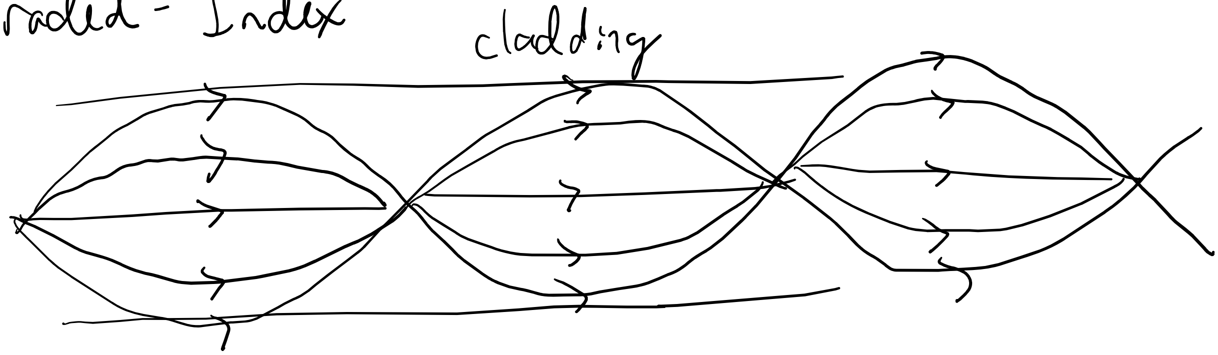
- Graded Index fiber (improves dispersion Reduction)



Step-Index



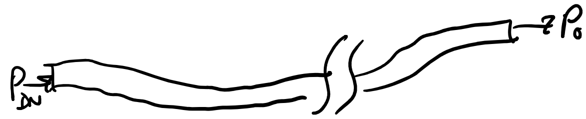
Graded-Index



Losses

Optical fiber attenuation (α) → due to absorption

$$\boxed{\alpha = \left(\frac{1}{L}\right) \ln\left(\frac{P_{IN}}{P_o}\right)}$$



$$\boxed{P_o = P_{In} e^{-\alpha L}}$$

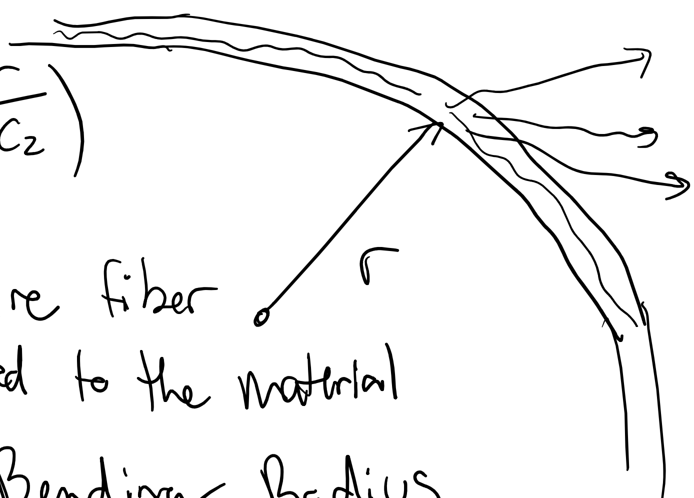
$$\boxed{\alpha_{dB} = \frac{10}{\ln(10)}(\alpha) \approx 4.34\alpha}$$

$$\boxed{P_{dBm} = 10 \log_{10}\left(\frac{P_{mW}}{1mW}\right)}$$

dBm is based on 1mW

Bending Loss

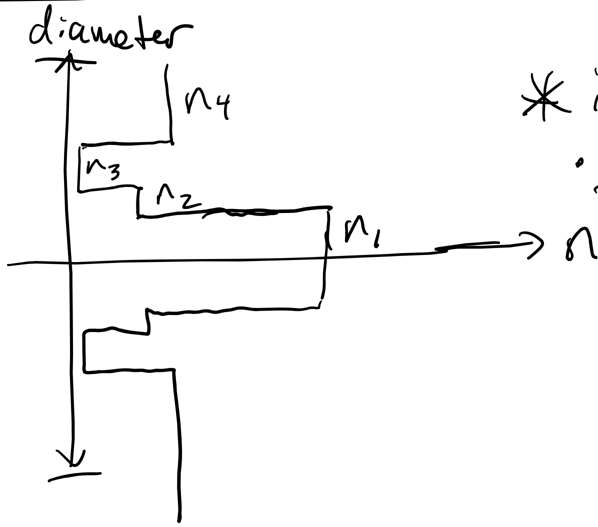
$$\alpha_B = C_1 e^{-\left(\frac{r}{C_2}\right)}$$



- C_1 and C_2 are fiber constants Related to the material
- r is the Bending Radius

- Tailor fiber to be more Resistent to bends

Trench Fiber Design



* improves bending
- allowing greater bending radius